

Cluster-Based Federated Learning for Diabetes Prediction: A Multi-Case Evaluation Across Diverse Datasets - MAX 25 PAGES

EDA BAHAR* and OZGUN PINARER*, Galatasaray University, Dept. Computer Engineering, Turkey

Machine learning (ML) can enhance healthcare by facilitating early diagnosis and personalized treatment, but its adoption is hindered by privacy concerns and the decentralized nature of medical data. Federated Learning (FL) addresses these issues by training models collaboratively without sharing raw patient data. Still, most FL methods do not fully account for differences between patient groups, which can limit their practical usefulness. In this study, we present a cluster-based federated learning (CFL) approach for diabetes prediction. Our method groups patients into clinically meaningful subgroups and then combines local models through federated aggregation. We test this approach on three different datasets: a large diabetes prediction dataset, the Pima Indians Diabetes dataset, and a CDC health indicators dataset. We compare Logistic Regression (LR) and Random Forest Classifier (RFC) models under centralized and federated settings. The results show that CFL achieves accuracy close to centralized models while protecting data privacy. RFC consistently performs better than LR, and clustering improves fairness and subgroup-specific learning, especially in diverse datasets. Our main contribution is providing the first systematic comparison of CFL across multiple diabetes datasets, offering practical guidance for privacy-preserving and personalized predictive modeling in healthcare.

CCS Concepts: • **Computing methodologies** → **Classification and regression trees**; *Neural networks*; **Cluster analysis**.

Additional Key Words and Phrases: xx,xx,xx

ACM Reference Format:

Eda Bahar and Ozgun Pinarer. 2018. Cluster-Based Federated Learning for Diabetes Prediction: A Multi-Case Evaluation Across Diverse Datasets - MAX 25 PAGES. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 18 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

The increasing availability of healthcare data, combined with rapid advances in ML, presents substantial opportunities to improve clinical decision-making, disease prediction, and patient care. Health institutions generate vast amounts of structured and unstructured data through electronic health records (EHRs), medical imaging, remote patient monitoring, and genomic studies. Harnessing these datasets with ML techniques can support early diagnosis, personalized treatment planning, and population-level health management. However, applying ML in healthcare is not straightforward. The sensitive nature of medical data, strict privacy regulations such as Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR), and the distributed nature of datasets across multiple institutions create significant barriers to effective data sharing. Traditional centralized ML approaches require pooling

*Both authors contributed equally to this research.

Authors' Contact Information: [Eda Bahar](mailto:ebahar@gsu.edu.tr), ebahar@gsu.edu.tr; [Ozgun Pinarer](mailto:opinarer@gsu.edu.tr), opinarer@gsu.edu.tr, Galatasaray University, Dept. Computer Engineering, Istanbul, Turkey.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

data into a single location for training, which not only risks breaching patient privacy but also limits collaboration and model generalizability across diverse populations.

To address these challenges, FL has emerged as a promising paradigm. It allows multiple institutions to collaboratively train predictive models without exchanging raw patient data. Only model parameters or updates are shared, which enables the development of accurate, robust, and generalizable models while preserving privacy. FL has been successfully applied to various healthcare domains, including EHR-based predictive modeling, disease risk assessment, medical imaging classification, and remote patient monitoring. Its decentralized nature also accommodates heterogeneous datasets, varying data quality, and differing institutional resources. Recent extensions of FL, such as CFL, further enhance its applicability by accounting for differences in data distribution and resource constraints. By combining patient subgrouping with decentralized aggregation, these approaches improve predictive performance, enable more personalized analytics, and ensure compliance with privacy regulations.

Despite these advancements, several gaps remain. Most existing studies focus on general FL frameworks without exploring clinically meaningful patient subgrouping or feature-based clustering. In addition, there is limited empirical evidence comparing federated approaches to traditional centralized models in terms of predictive accuracy and practical utility, particularly for chronic disease prediction such as diabetes. Understanding how subgroup-aware federated approaches perform across multiple, heterogeneous datasets is crucial for both clinical adoption and methodological advancement.

In this work, we aim to address these gaps by integrating feature-based clustering with federated learning for diabetes prediction. Our study examines the formation of clinically relevant patient subgroups, evaluates model performance under decentralized training conditions, and compares federated aggregation against conventional centralized approaches. By doing so, we provide a systematic analysis of the potential benefits of FL in improving predictive accuracy, supporting personalized healthcare, and preserving patient privacy, thereby contributing both methodological insights and practical guidance for deploying ML in distributed healthcare environments.

The remainder of the paper is organized as follows: Detailed literature review is introduced in Section 2. Section 3 presents the proposed approach and methodology of this study. Section 4 describes experimental environments, metrics and performed experiments in detail. Finally, conclusion and discussion are presented in Section 5.

2 Related Work

FL has emerged as a transformative approach in healthcare, enabling robust, accurate, and privacy-preserving ML while addressing the challenges posed by fragmented medical data and strict privacy regulations. Rieke et al. [22] highlight that traditional centralized approaches are limited by “data silos” and privacy constraints, preventing the full exploitation of large-scale datasets. FL overcomes these challenges by allowing collaborative model training across multiple institutions without sharing raw patient data, exchanging only model updates. This paradigm supports the creation of generalizable Artificial Intelligence (AI) models and advances precision medicine.

Surveys and systematic reviews provide a comprehensive understanding of FL in healthcare. Nguyen et al. [18] categorize FL into horizontal, vertical, and federated transfer learning, emphasizing its advantages over centralized models, such as enhanced privacy and reduced training costs. They examine advanced frameworks, including resource-aware, secure, privacy-preserving, incentive-driven, and personalized designs, and their applications in EHR management, remote health monitoring, medical imaging, and COVID-19 diagnosis, while outlining open challenges in communication efficiency, standardization, data quality, security, and integration with next-generation networks. Antunes et al. [1] focus on architectures for EHRs, aggregation strategies, privacy mechanisms, and federated data analysis, proposing a

comprehensive framework for secure FL deployment. Prayitno et al.[21] further highlight key challenges, including uneven data distribution, data partitioning, and the need for strong privacy-preserving techniques such as differential privacy and homomorphic encryption. Xu et al.[27] investigate FL in healthcare informatics, addressing statistical and communication challenges, and exploring applications such as predictive modeling and patient similarity analysis. Cremonesi et al. [7] focus on multimodal health data for rare diseases, advocating for a Common Data Model (CDM) to enable collaborative training while avoiding centralization, and stressing the importance of harmonized standards across clinical and omics domains.

Practical applications of FL demonstrate its effectiveness in privacy-preserving medical analytics. Xu et al.[28] introduce FedMood for mood disorder detection using mobile health data, achieving high accuracy despite heterogeneous inputs. Brisimi et al.[4] develop the Cluster Primal Dual Splitting (cPDS) algorithm for predicting cardiac-related hospitalizations across distributed EHRs, improving convergence speed and reducing communication overhead. Dayan et al.[9] present EXAM, a federated model predicting oxygen needs in COVID-19 patients, demonstrating improved accuracy and generalizability across 20 global sites. Elayan et al.[12] and Sun et al.[25] extend FL to IoT-based healthcare monitoring, enabling decentralized model training on edge devices with high predictive performance and low operational cost. Guo et al.[13] propose a two-stage incremental learning strategy for real-time medical data, preserving prior knowledge while reducing computational overhead.

Additional systematic reviews [10, 20] stress FL's role in protecting sensitive medical data, discussing privacy-preserving techniques such as differential privacy, k-anonymity, l-diversity, and confidential computing. Bahar et al.[2] provide a structured overview of FL, including Resource-Constrained FL, emphasizing its applications in resource-limited environments like IoT devices and edge networks, and identifying research gaps for future development. Bashir et al.[3] examine the integration of FL with the Healthcare Metaverse, highlighting applications in medical diagnosis, patient monitoring, collaborative research, and drug discovery, while considering future challenges such as limited computing resources and the need for explainable AI. Chaddad et al.[5] analyze FL applications in medical imaging, disease diagnosis, and Magnetic Resonance Imaging (MRI) reconstruction, and discuss challenges including data heterogeneity, malicious attacks, and the lack of unified benchmarks. Joshi et al.[16] survey FL architectures, aggregation algorithms, and partitioning strategies, showing how FL improves collaborative healthcare applications while safeguarding patient privacy. Dang et al. [8] evaluate FL for EHR-based tasks such as in-hospital mortality and acute kidney injury prediction, demonstrating performance comparable to centralized approaches while preserving privacy.

CFL has emerged as a solution to address data heterogeneity and enhance personalized model performance. Zhou et al.[30] introduce a hierarchical FL framework for Internet of Medical Things (IoMT) applications, clustering participants by social context and aggregating models across edge and cloud layers. Cheema et al.[6] propose CFL for structural health monitoring of bridges, clustering data based on similarity for efficient cluster-specific training. Jiang et al.[15] apply CFL to personalized stress monitoring, employing a cold-start algorithm to integrate new clients without extensive retraining. Kandati and Gadekallu [17] develop Genetic CFL for early COVID-19 detection, combining FL with a genetic algorithm for cluster-based hyperparameter optimization, thereby improving accuracy and computational efficiency.

Collectively, these studies demonstrate the potential of FL and CFL for scalable, privacy-preserving, and efficient learning across heterogeneous healthcare datasets. This body of work provides the foundation for the CFL methodology presented in the following section.

3 Proposed Approach and Methodology

In this study, we aim to (burayi yazamadim)

3.1 Dataset

For this study, the first dataset is the Diabetes Prediction Dataset, available on Kaggle (iammustafatz). The dataset initially consists of 100,000 patient records, each described by multiple demographic, lifestyle, and clinical features. The features include age, gender, body mass index (BMI), HbA1c level, blood glucose level, smoking history, and other relevant health indicators. The target variable is a binary label indicating whether a patient has diabetes. Prior to analysis, the dataset was carefully preprocessed to ensure quality and consistency. Missing data were handled either by imputation or by assigning “No Info” categories where medical measurements were unavailable. Categorical variables such as gender and smoking history were encoded into numeric representations suitable for machine learning models. [29]

The second dataset used in this study is the Diabetes Dataset by Akshay Dattatray Khare, which is available on Kaggle. It is derived from the well-known Pima Indians Diabetes Study, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases. This dataset has become a benchmark resource in the field of machine learning for evaluating predictive models in medical research, particularly for the task of diabetes classification.

The dataset consists of eight diagnostic and demographic features that serve as predictors. These include the number of pregnancies, plasma glucose concentration, diastolic blood pressure, triceps skinfold thickness, 2-hour serum insulin, BMI, diabetes pedigree function, and age. The target variable, referred to as Outcome, indicates whether an individual has diabetes, with a value of 1 representing a positive diagnosis and 0 representing a negative one.

Because it combines biochemical measurements with demographic information, the dataset is clinically meaningful and offers a compact yet comprehensive basis for prediction. It is frequently used in research as a benchmark for supervised learning models due to its accessibility, structured nature, and clear medical relevance. Moreover, its frequent citation in literature underscores its role as a reliable dataset for assessing the performance of classification algorithms in healthcare applications.

Finally, the third dataset used in this study is the Diabetes Health Indicators dataset, specifically the balanced version titled `diabetes_binary_5050split_health_indicators_BRFSS2015.csv`, which contains 70,692 survey responses from the U.S. Centers for Disease Control and Prevention’s (CDC) Behavioral Risk Factor Surveillance System (BRFSS) 2015 survey. [11] This dataset, prepared and shared on Kaggle by Alex Teboul, is designed to provide a cleaned and balanced sample for practical machine learning applications.

The dataset includes 21 features spanning health, lifestyle, and demographic indicators. Among the most prominent predictors are BMI, high blood pressure, high cholesterol, smoking status, alcohol consumption, physical activity, fruit and vegetable intake, and general health rating. Additionally, it incorporates demographic variables such as age (categorized in intervals), education level, and income. The target variable, `Diabetes_binary`, distinguishes individuals with diabetes (value of 1) from those without diabetes (value of 0).

By balancing positive and negative cases of diabetes, this dataset provides an improved foundation for supervised learning, thereby reducing the bias introduced by class imbalance in large-scale health survey data. Its breadth of behavioral and demographic variables, combined with biomedical indicators, makes it especially suitable for studying multifactorial influences on diabetes risk. Furthermore, its structured and preprocessed nature facilitates reproducible experimentation in federated and clustered learning scenarios.

3.2 Feature-Based Clustering

To explore heterogeneity among patients, clustering was performed based on clinically meaningful feature categories. The first clustering strategy was built on BMI, which was divided into medically recognized categories: underweight (less than 18.5), healthy weight (18.5 to less than 25), overweight (25 to less than 30), class I obesity (30 to less than 35), class II obesity (35 to less than 40), class III obesity or severe obesity (40 or greater), and an additional “No Info” category for missing values. [26] The second strategy relied on HbA1c levels, which are widely used for diagnosing diabetes. Patients were categorized as normal (less than 5.7), pre-diabetes (5.7 to less than 6.5), diabetes (6.5 or greater), or “No Info”. [23] A third clustering strategy was defined by fasting blood glucose levels, categorized into normal (70 to 99 mg/dL), pre-diabetes (100 to 125 mg/dL), diabetes (126 mg/dL or greater), and “No Info”. [24] A fourth clustering strategy was the gender categories: male, female and other (not specified). The next clustering strategies were age, hypertension, whether they have heart disease or not, difficulty at walking, physical health, general health, high cholesterol, blood pressure, insulin level, pregnancies, skin thickness, and diabetes pedigree function. Finally, smoking history was used as another categorical clustering feature, capturing the lifestyle component of diabetes risk. Each of these clustering approaches was motivated by its strong clinical relevance in diabetes diagnosis and prevention, enabling the creation of patient subgroups with distinct risk profiles.

In addition to these clinically defined bins, unsupervised clustering was applied using k-means to identify data-driven patient subgroups based on each continuous feature. The optimal number of clusters was determined using the silhouette score, ensuring that intra-cluster similarity was maximized while maintaining high inter-cluster separation. [19] This combination of clinical thresholds and k-means clustering allowed for both interpretable and statistically robust subgroup formation, providing meaningful partitions for subsequent federated learning analyses. Each clustering approach was motivated by strong clinical relevance in diabetes diagnosis and prevention, enabling the creation of patient subgroups with distinct risk profiles.

3.3 Models

For the classification task, two commonly used models were employed. LR was selected as a baseline linear model due to its interpretability and clinical transparency, which allows for a direct assessment of how each feature contributes to diabetes risk. RFC classifier was chosen as a complementary non-linear model, as it can capture complex interactions among features while being robust against noise.

To explore the benefits of FL, separate models were trained on each patient subgroup, reflecting the decentralized nature of this approach. In the case of LR, federated aggregation was achieved by weighting the cluster-specific coefficients according to their sample sizes. For the RFC model, federated aggregation involved averaging the predicted probabilities from each cluster’s model, effectively simulating a distributed consensus.

This approach enabled a comparison between feature-based clustering in FL and traditional centralized training, while maintaining subgroup-specific learning.

3.4 Evaluation Metrics

To assess the performance of the models, four widely used classification metrics were employed: accuracy, precision, recall, and f1-score. [14] Accuracy provides an overall measure of how often the models correctly classified individuals, while precision reflects the proportion of true positive predictions among all positive predictions. Recall, on the other hand, measures the ability of the models to identify positive cases correctly, which is particularly important in medical

Table 1. Datasets Description

Dataset	Record	Number of Features
Dataset 1	100000	8
Dataset 2	768	8
Dataset 3	70692	21

contexts where missing a diagnosis can have serious consequences. The f1-score combines precision and recall into a single harmonic mean, offering a balanced metric when there is a trade-off between the two.

The evaluation was performed at two levels. First, the performance of federated learning models trained within each cluster was measured using these metrics, allowing for the assessment of how well models tailored to specific subgroups captured their unique patterns. Second, the global performance of the federated model was evaluated by aggregating results across clusters, allowing for a comparison with models trained on centralized data without clustering. This approach provided insights into both the effectiveness of clustering strategies and the added value of federated learning in preserving predictive performance while respecting subgroup distinctions.

4 Experimentation

Table 1 summarizes the three datasets used in this study. It provides the number of records and the number of features for each dataset. Dataset 1 contains 100,000 records with 8 features and is available on Kaggle. Dataset 2 includes 768 records with 8 features and is derived from the Pima Indians Diabetes Study. Dataset 3 has 70,692 records with 21 features and consists of health indicators collected from the CDC’s BRFSS2015 survey. The table also includes hyperlinks to the original sources for reference.

In all clustering experiments, clusters were generated using the K-means algorithm. The optimal number of clusters was determined with the silhouette score, which measures how well each instance fits within its assigned cluster compared to other clusters. This approach ensured that the partitions were both meaningful and consistent across different feature-based groupings.

4.1 Global vs Aggregated Model Performance on Dataset 1

Table 2 summarizes the performance of both centralized global models and FL models aggregated across different clustering strategies for diabetes prediction. Overall, the RFC classifier consistently outperformed LR across all clusters in terms of accuracy, precision, recall, and f1-score.

The global LR model achieved an accuracy of 0.9571, a precision of 0.8545, a recall of 0.6281, and an f1-score of 0.7240. By contrast, the global RFC attained higher performance, with an accuracy of 0.9683 and notably stronger precision (0.9469) and f1-score (0.7950). Similar trends were observed when models were trained and aggregated across clusters. For example, when clustering by age, the federated LR model achieved an accuracy of 0.9576 and an f1-score of 0.7289, while the corresponding RFC achieved an accuracy of 0.9707 and an f1-score of 0.8044.

Clustering by gender, hypertension, heart disease, and BMI yielded comparable results, with aggregated RFC models achieving accuracies above 0.9700 and f1-scores exceeding 0.802, whereas LR models showed lower recall (ranging between 0.5956 and 0.6368) despite maintaining high precision. Interestingly, clustering by HbA1c levels resulted in the weakest recall for LR (0.5003), reflecting difficulties in capturing the heterogeneity of this subgroup, although its

Table 2. Performance of Global Models and Aggregated FL Models for Diabetes Classification (Dataset 1)

Feature	Model	Metrics			
		Accuracy	Precision	Recall	f1-score
Global	LR	0.9571	0.8545	0.6281	0.7240
	RFC Classifier	0.9683	0.9469	0.6850	0.7950
Gender	LR	0.9574	0.8568	0.6292	0.7256
	RFC Classifier	0.9700	0.9798	0.6798	0.8027
Age	LR	0.9576	0.8522	0.6368	0.7289
	RFC Classifier	0.9707	1.0000	0.6728	0.8044
Hypertension	LR	0.9572	0.8692	0.6141	0.7197
	RFC Classifier	0.9704	0.9800	0.6833	0.80524
Heart Disease	LR	0.9566	0.8602	0.6153	0.7174
	RFC Classifier	0.9704	0.9956	0.6722	0.8026
BMI	LR	0.9581	0.9039	0.5956	0.7180
	RFC Classifier	0.9706	1.0000	0.6711	0.8032
HbA1c Level	LR	0.9542	0.9773	0.5003	0.6618
	RFC Classifier	0.9666	0.9607	0.6537	0.7780
Blood Glucose Level	LR	0.9569	0.8519	0.6281	0.7231
	RFC Classifier	0.9659	0.9087	0.6886	0.7835

precision remained high (0.9773). The aggregated RFC model, however, preserved a balanced performance with an f1-score of 0.7780 under this clustering.

Taken together, these results demonstrate that federated RFC models consistently achieved superior predictive balance compared to LR, regardless of the clustering strategy applied. While clustering did not significantly increase global performance relative to centralized models, it maintained comparable predictive accuracy while enabling subgroup-specific training. This finding highlights the potential of feature-based clustering in federated contexts, where subgroup diversity can be leveraged without sacrificing overall classification performance. When comparing centralized global learning to federated learning, the differences in raw predictive performance were not substantial. In most cases, the federated models performed on par with, or slightly below, their centralized counterparts. However, the advantage of the federated approach lies in its ability to train robust models without requiring data centralization, thus respecting privacy and addressing data governance challenges. Moreover, clustering introduced flexibility to adapt model training to population subgroups, offering a pathway toward fairness and personalization in diabetes risk prediction.

4.2 Local Model Performances of Clusters of Dataset 1

Table 3 and table 4 present the performance of LR and RFC models trained locally within each cluster for the diabetes prediction task. Across most feature-based clusters, RFC consistently outperformed LR, achieving higher accuracy, precision, and f1-scores while maintaining competitive recall.

For gender-based clusters, the sizes were 56,161 for Female, 39,967 for Male, and 18 for Other. RFC achieved accuracies of 0.9725, 0.9666, and 1.0000 for Female, Male, and Other, respectively, while LR ranged from 0.9530 to 1.0000. The “Other” category contained only one class in the training data, which resulted in perfect accuracy but undefined precision, recall, and f1-score.

Age-based clusters contained 32,449 instances in cluster 0 (mean age 66.87), 28,572 in cluster 1 (mean age 14.24), and 35,125 in cluster 2 (mean age 41.03). Performance varied across these clusters, with extreme or underrepresented age groups showing lower recall for LR, while RFC maintained robust f1-scores above 0.79.

For BMI-based clusters, the record counts were: Underweight (<18.5) 8,491, Healthy Weight (18.5–24.9) 22,208, Overweight (25–29.9) 41,917, Class 1 Obesity (30–34.9) 12,732, Class 2 Obesity (35–39.9) 6,177, and Class 3 Obesity (≥ 40) 4,621. LR showed weaker recall in underweight and less frequent BMI categories, whereas RFC consistently achieved f1-scores above 0.79, demonstrating strong predictive capability even in smaller clusters.

For metabolic markers, HbA1c clusters included Normal (<5.7, 36,265), Pre-diabetes (5.7–6.4, 39,778), and Diabetes (≥ 6.5 , 20,103), and glucose clusters included Diabetes (≥ 126 , 69,132), Normal (70–99, 20,240), and Pre-diabetes (100–125, 6,774). Clusters such as HbA1c Normal and glucose Normal or Pre-diabetes contained only a single target class in the training data, producing perfect accuracy but undefined precision, recall, and f1-scores for minority class detection. RFC outperformed LR in heterogeneous clusters, capturing non-linear patterns effectively.

Clusters based on Hypertension (0: 88,685; 1: 7,461), Heart Disease (0: 92,223; 1: 3,923), and Smoking History (Never: 34,398; No Info: 32,887; Former: 9,299; Current: 9,197; Not Current: 6,367; Ever: 3,998) similarly demonstrated that RFC handled class imbalance better, while LR struggled in smaller or highly skewed clusters.

In general, these results highlight the benefits of cluster-specific training in a federated learning setting. RFC models are more resilient to class imbalance within clusters, whereas LR may fail to identify minority class instances when clusters are small or homogeneous. These findings suggest that federated approaches using feature-based clustering can yield accurate local models, but careful selection and monitoring of clusters are essential to avoid trivial solutions in sparse or single-class subgroups.

When comparing local cluster models to the aggregated FL models reported in Table 2, it is evident that aggregation provides a more balanced global performance. While individual clusters occasionally exhibited very high accuracy due to single-class dominance, the aggregated LR and RFC models achieve consistently high performance across the entire test set, mitigating the skew and variance observed in smaller or unbalanced clusters. For instance, although some HbA1c and blood glucose clusters showed perfect accuracy with zero f1-score locally, the corresponding aggregated models produced balanced f1-scores of 0.6618–0.7950, reflecting the combined knowledge from all clusters.

Furthermore, feature-based clustering allowed the FL models to capture heterogeneity in the patient population, improving predictive robustness compared to a purely centralized model.

4.3 Global vs Aggregated Model Performance on Dataset 2

The performance of both global models and aggregated FL models for diabetes classification on Dataset 2 is summarized in Table 5. Overall, the global LR model achieved an accuracy of 0.7468, with a precision of 0.6379, a recall of 0.6727, and an f1-score of 0.6549. The global RFC model slightly outperformed LR in terms of accuracy, achieving a value of 0.7662. However, precision (0.6863) and recall (0.6364) reflected a trade-off typical of tree-based models, resulting in an f1-score of 0.6604.

When the dataset was partitioned based on individual features for FL, variations in performance were observed across different clusters. Clustering by pregnancies yielded identical performance for both models, suggesting that this feature alone has limited discriminatory power. In contrast, clustering by glucose levels highlighted differences between the two models: LR achieved an accuracy of 0.7662 and a balanced f1-score of 0.6170, while the RFC displayed high precision (0.8889) but very low recall (0.1455), resulting in a poor f1-score of 0.2500. This indicates that the RFC over-predicted the negative class in some clusters, leading to class imbalance issues.

Table 3. Performance of Local Models on Clusters for Diabetes Classification (Dataset 1)

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
Gender	Female	LR	0.9605	0.8538	0.6152	0.7150
		RFC Classifier	0.9725	0.9821	0.6718	0.7978
	Male	LR	0.9530	0.8599	0.6446	0.7369
		RFC Classifier	0.9666	0.9774	0.6886	0.8080
	Other	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	1.0000	0.0000	0.0000	0.0000
Age	0	LR	0.9131	0.8377	0.6728	0.7462
		RFC Classifier	0.9377	1.0000	0.6720	0.8038
	1	LR	0.9956	0.9411	0.4000	0.6140
		RFC Classifier	0.9977	1.0000	0.6750	0.8059
	2	LR	0.9678	0.8989	0.5582	0.6887
		RFC Classifier	0.9792	1.0000	0.6748	0.8058
Hypertension	0	LR	0.9630	0.8557	0.5953	0.7021
		RFC Classifier	0.9758	0.9800	0.6833	0.8052
	1	LR	0.8897	0.9079	0.6713	0.7719
		RFC Classifier	0.9080	0.9797	0.6830	0.8049
Heart Disease	0	LR	0.9634	0.8633	0.5953	0.7047
		RFC Classifier	0.9755	0.9778	0.6810	0.8029
	1	LR	0.8969	0.9466	0.6666	0.7823
		RFC Classifier	0.9112	0.9898	0.6877	0.8116
Smoking History	No Info	LR	0.9768	0.7904	0.6080	0.6873
		RFC Classifier	0.9864	0.9946	0.6813	0.8086
	never	LR	0.9549	0.8936	0.6149	0.7285
		RFC Classifier	0.9684	0.9957	0.6822	0.8097
	former	LR	0.9140	0.8319	0.6226	0.7122
		RFC Classifier	0.9328	0.9898	0.6132	0.7572
	current	LR	0.9539	0.9090	0.6010	0.7236
		RFC Classifier	0.9726	1.0000	0.7267	0.8417
	not current	LR	0.9458	0.8793	0.6415	0.7418
		RFC Classifier	0.9611	1.0000	0.6792	0.8089
	ever	LR	0.9288	0.8289	0.6000	0.6961
		RFC Classifier	0.9534	1.0000	0.6571	0.7931

Other features, such as insulin levels, blood pressure, skin thickness, and diabetes pedigree function, yielded moderate to strong performance for both models. For example, clustering by insulin yielded an accuracy of 0.7848 for LR and 0.7595 for RFC, with reasonably balanced f1-scores of 0.6531 and 0.5581, respectively. Notably, clustering by BMI led to a lower overall performance for both models, particularly for the RFC, which showed high precision (0.7500) but extremely low recall (0.2727), resulting in an f1-score of 0.4000. This indicates that certain BMI clusters may contain highly imbalanced class distributions.

Overall, LR demonstrated more consistent performance across clusters, while RFC occasionally achieved perfect or near-perfect precision at the expense of recall. These results demonstrate the impact of feature-based clustering on federated learning performance, highlighting that features with strong clinical relevance, such as glucose and insulin,

Table 4. Performance of Local Models on Clusters for Diabetes Classification (Dataset 1) ctd.

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
BMI	Underweight (<18.5)	LR	0.9958	1.0000	0.4167	0.5882
		RFC Classifier	0.9982	1.0000	0.7500	0.8571
	Healthy Weight (18.5-24.9)	LR	0.9799	0.8932	0.5412	0.6740
		RFC Classifier	0.9869	1.0000	0.6588	0.7943
	Overweight (25-29.9)	LR	0.9615	0.9261	0.5795	0.7129
		RFC Classifier	0.9731	1.0000	0.6734	0.8048
	Class 1 Obesity (30-34.9)	LR	0.9310	0.8884	0.6011	0.7170
		RFC Classifier	0.9510	1.0000	0.6631	0.7974
	Class 2 Obesity (35-39.9)	LR	0.9251	0.8941	0.6726	0.7677
		RFC Classifier	0.9520	1.0000	0.7389	0.8499
	Class 3 Obesity (>=40)	LR	0.8768	0.8837	0.6080	0.7204
		RFC Classifier	0.9008	1.0000	0.6200	0.7654
HbA1c Level	Normal (<5.7)	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	0.9963	0.0000	0.0000	0.0000
	Pre-diabetes (5.7-6.4)	LR	0.9258	0.9881	0.1233	0.2193
		RFC Classifier	0.9490	0.9406	0.4235	0.5840
	Diabetes (>=6.5)	LR	0.9275	0.9762	0.7424	0.8434
		RFC Classifier	0.9476	0.9988	0.8015	0.8894
Blood Glucose Level	Normal (70-99)	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	0.9930	0.0000	0.0000	0.0000
	Pre-diabetes (100-125)	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	0.9925	0.0000	0.0000	0.0000
	Diabetes (>=126)	LR	0.9404	0.8519	0.6281	0.7231
		RFC Classifier	0.9556	0.9360	0.6886	0.7934

yield more informative clusters for diabetes prediction. In contrast, features with weaker predictive power or imbalanced distributions, like BMI, may introduce challenges to model generalization.

When comparing the performance across the two datasets, notable differences emerge in both global and federated learning models. Dataset 1, with a larger sample size and a more comprehensive set of features, including HbA1c and blood glucose levels, generally yielded higher accuracies and f1-scores for both LR and RFC models, with aggregated FL models achieving performance close to that of centralized global models. In contrast, Dataset 2, which is smaller and includes fewer features, displayed lower overall predictive performance, particularly for RFC models in certain clusters, such as glucose and BMI, where high precision was accompanied by low recall, resulting in imbalanced f1-scores. LR exhibited more stable performance across clusters in both datasets, reflecting its robustness in smaller or less heterogeneous cohorts. These observations suggest that feature richness, sample size, and clinically meaningful clustering have a significant influence on the effectiveness of federated learning. That federated aggregation benefits more when clusters are well-separated and informative, as seen in Dataset 1.

4.4 Local Model Performances of Clusters of Dataset 2

The performance of local models trained on clusters for Dataset 2 in table ?? and table ?? show substantial variability across features and clusters, reflecting the heterogeneous nature of the dataset.

Table 5. Performance of Global Models and Aggregated FL Models for Diabetes Classification (Dataset 2)

Feature	Model	Metrics			
		Accuracy	Precision	Recall	f1-score
Global	LR	0.7468	0.6379	0.6727	0.6549
	RFC Classifier	0.7662	0.6863	0.6364	0.6604
Pregnancies	LR	0.7338	0.6167	0.6727	0.6435
	RFC Classifier	0.7338	0.6167	0.6727	0.6435
Glucose	LR	0.7662	0.7436	0.5273	0.6170
	RFC Classifier	0.6883	0.8889	0.1455	0.2500
Insulin	LR	0.7848	0.6667	0.6400	0.6531
	RFC Classifier	0.7595	0.6667	0.4800	0.5581
Blood Pressure	LR	0.7792	0.7333	0.6000	0.6600
	RFC Classifier	0.7662	0.6462	0.7636	0.7000
BMI	LR	0.6818	0.5417	0.7091	0.6142
	RFC Classifier	0.7078	0.7500	0.2727	0.4000
Skin Thickness	LR	0.7615	0.6591	0.7250	0.6905
	RFC Classifier	0.7615	0.7188	0.5750	0.6389
Diabetes Pedigree Function	LR	0.7468	0.6379	0.6727	0.6549
	RFC Classifier	0.7273	0.6000	0.7091	0.6500
Age	LR	0.7143	0.5821	0.7091	0.6393
	RFC Classifier	0.7273	0.6102	0.6545	0.6316

For pregnancies-based clusters, there were 492 records in cluster 0 (mean 1.70 pregnancies) and 276 records in cluster 1 (mean 7.66 pregnancies). LR achieved accuracies ranging from 0.667 to 0.777, while RFC performed slightly better in some clusters, reaching 0.700. Very small clusters, such as rarely represented categories or those containing only a single class, occasionally produced perfect accuracy (1.0000) but zero precision, recall, and f1-score, indicating that meaningful evaluation was impossible.

Glucose Level clusters contained 181 records for Normal (70–99 mg/dL), 297 for Diabetes (≥ 126 mg/dL), 274 for Pre-diabetes (100–125 mg/dL), and 16 for No Info. LR and RFC models performed well in the Diabetes cluster, with LR reaching an f1-score of 0.767. Other clusters, including Normal and Pre-diabetes, showed low or zero recall and precision due to class imbalance, and the No Info cluster contained only one class, leading to trivial predictions.

Blood pressure clusters included 730 records for 0 (mean 730) and 38 records for 1 (mean 2.21), while Insulin clusters had 338 records for 0 (mean 388.21) and 56 for 1 (mean 117.00). Both features showed moderate to high predictive performance. LR generally achieved slightly higher f1-scores, whereas RFC maintained balanced performance across clusters.

BMI-based clusters demonstrated notable differences. The counts were: Underweight (<18.5) 4 (single-class), Healthy Weight (18.5–24.9) 102, Overweight (25–29.9) 179, Class 1 Obesity (30–34.9) 224, Class 2 Obesity (35–39.9) 150, Class 3 Obesity (40) 98, and No Info 11 (single-class). LR sometimes reached perfect accuracy in small clusters such as Healthy Weight or single-class clusters, but this was accompanied by zero precision and recall when only one target class was present. Class 2 and 3 obesity clusters had lower accuracy but higher recall, capturing more positive cases at the expense of false positives. RFC maintained high recall in most clusters but occasionally had imbalanced f1-scores due to low precision in small clusters.

Table 6. Performance of Local Models on Clusters for Diabetes Classification (Dataset 2)

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
Pregnancies	0	LR	0.7766	0.6071	0.6296	0.6182
		RFC Classifier	0.7553	0.5667	0.6296	0.5965
	1	LR	0.6667	0.6250	0.7143	0.6667
		RFC Classifier	0.7000	0.6667	0.7143	0.6897
Glucose Level	No Info	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	1.0000	0.0000	0.0000	0.0000
	Normal (70-99)	LR	0.9048	0.0000	0.0000	0.0000
		RFC Classifier	0.9048	0.0000	0.0000	0.0000
	Pre-diabetes (100-125)	LR	0.7000	0.2500	0.0769	0.1176
		RFC Classifier	0.7600	1.0000	0.0769	0.1429
	Diabetes (≥ 126)	LR	0.7167	0.8000	0.7368	0.7671
		RFC Classifier	0.4667	0.8750	0.1842	0.3043
Insulin	0	LR	0.7857	0.7273	1.0000	0.8421
		RFC Classifier	0.7143	0.8333	0.6250	0.7143
	1	LR	0.7846	0.6154	0.4706	0.5333
		RFC Classifier	0.7692	0.5833	0.4118	0.4828
Blood Pressure	0	LR	0.7832	0.7368	0.5714	0.6437
		RFC Classifier	0.7762	0.6441	0.7755	0.7037
	1	LR	0.7273	0.7143	0.8333	0.7692
		RFC Classifier	0.6364	0.6667	0.6667	0.6667

Skin Thickness clusters included 242 records in cluster 0 (mean 30.71), 184 in cluster 1 (mean 18.03), and 115 in cluster 2 (mean 43.64), while Diabetes pedigree function clusters had 194 records in cluster 0 (mean 0.92) and 574 in cluster 1 (mean 0.31). LR tended to perform better in smaller clusters, whereas RFC provided competitive f1-scores in larger clusters.

Age-based clusters contained 533 records in cluster 0 (mean 26.60) and 235 in cluster 1 (mean 48.29). LR showed lower accuracy in younger clusters, reflecting more heterogeneous outcomes, while RFC maintained balanced performance, demonstrating robustness to complex interactions.

Overall, these results emphasize that local model performance is highly sensitive to cluster size and class distribution. Clusters with very few samples or single-class outcomes can produce misleadingly high accuracy, highlighting the importance of careful interpretation in federated learning. Clinically meaningful feature-based clustering reveals heterogeneous predictive patterns, which aggregated FL models can leverage to improve overall generalization.

4.5 Global vs Aggregated Model Performance on Dataset 3

Before conducting the experiments, duplicate rows were removed from the dataset (1635 entries). The target variable, Diabetes_binary, has two classes: 0 for no diabetes and 1 for pre-diabetes or diabetes. The dataset includes 21 feature variables in total.

Since clustering all 21 features would be inefficient and potentially dilute model performance, feature selection was applied based on correlation with the target variable as shown in Figure 1. Only features with a correlation greater than 0.20 with Diabetes_binary were considered for clustering. The selected features were: Age, Difficulty Walking,

Table 7. Performance of Local Models on Clusters for Diabetes Classification (Dataset 2) ctd.

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
BMI	No Info	LR	0.7500	0.0000	0.0000	0.0000
		RFC Classifier	0.7500	0.0000	0.0000	0.0000
	Underweight (<18.5)	LR	1.0000	0.0000	0.0000	0.0000
		RFC Classifier	1.0000	0.0000	0.0000	0.0000
	Healthy Weight (18.5-24.9)	LR	1.0000	1.0000	1.0000	1.0000
		RFC Classifier	1.0000	1.0000	1.0000	1.0000
	Overweight (25-29.9)	LR	0.6061	0.1667	0.1111	0.1333
		RFC Classifier	0.6667	0.0000	0.0000	0.0000
	Class 1 Obesity (30-34.9)	LR	0.6977	0.6818	0.7143	0.6977
		RFC Classifier	0.6279	0.7778	0.3333	0.4667
Skin Thickness	Class 2 Obesity (35-39.9)	LR	0.6176	0.4545	0.9091	0.6061
		RFC Classifier	0.7941	1.0000	0.3636	0.5333
	Class 3 Obesity (>=40)	LR	0.5714	0.5714	1.0000	0.7273
		RFC Classifier	0.5238	0.7500	0.2500	0.3750
	0	LR	0.7045	0.7000	0.6667	0.6829
		RFC Classifier	0.6364	0.6471	0.5238	0.5789
Diabetes Pedigree Function	1	LR	0.9231	1.0000	0.6667	0.8000
		RFC Classifier	0.8974	1.0000	0.5556	0.7143
	2	LR	0.6154	0.5000	0.9000	0.6429
		RFC Classifier	0.7692	0.7000	0.7000	0.7000
Age	0	LR	0.7209	0.7391	0.7391	0.7391
		RFC Classifier	0.7674	0.7826	0.7826	0.7826
	1	LR	0.7568	0.5714	0.6250	0.5970
		RFC Classifier	0.7117	0.5000	0.6562	0.5676
	0	LR	0.8081	0.7407	0.6250	0.6780
		RFC Classifier	0.7576	0.6333	0.5938	0.6129
	1	LR	0.5455	0.4750	0.8261	0.6032
		RFC Classifier	0.6727	0.5862	0.7391	0.6538

Physical Health (PhysHlth), General Health (GenHlth), Heart Disease or Attack, BMI, High Cholesterol, and High Blood Pressure.

Table 8 presents the performance comparison between global models and feature-based aggregated FL models for Dataset 3. Two classifiers, LR and RFC, were evaluated across accuracy, precision, recall, and f1-score. The global models, trained on the full dataset without feature-based partitioning, achieved accuracies of 0.7405 and 0.7482 for LR and RFC, respectively, with balanced precision and recall. These results establish the baseline performance. Among the aggregated FL models, several features yielded performance comparable to or better than the global baseline. For instance, the BMI-based LR model achieved the highest recall (0.8213) and f1-score (0.8258), surpassing the global LR model. Similarly, Difficulty Walking and Heart Disease or Attack features also showed strong results, particularly with RFC, which achieved recall values of 0.8101 and 0.8950, respectively. On the other hand, some features displayed weaker predictive power. The GenHlth variable, when modeled with LR, performed poorly across all metrics, with an accuracy of only 0.4992, reflecting its limited discriminatory capability. In contrast, RFC demonstrated greater robustness on the same feature, reaching an accuracy of 0.7344. Overall, the results highlight that federated aggregation on selected health indicators can achieve performance comparable to global training, with specific features—such as BMI, Difficulty

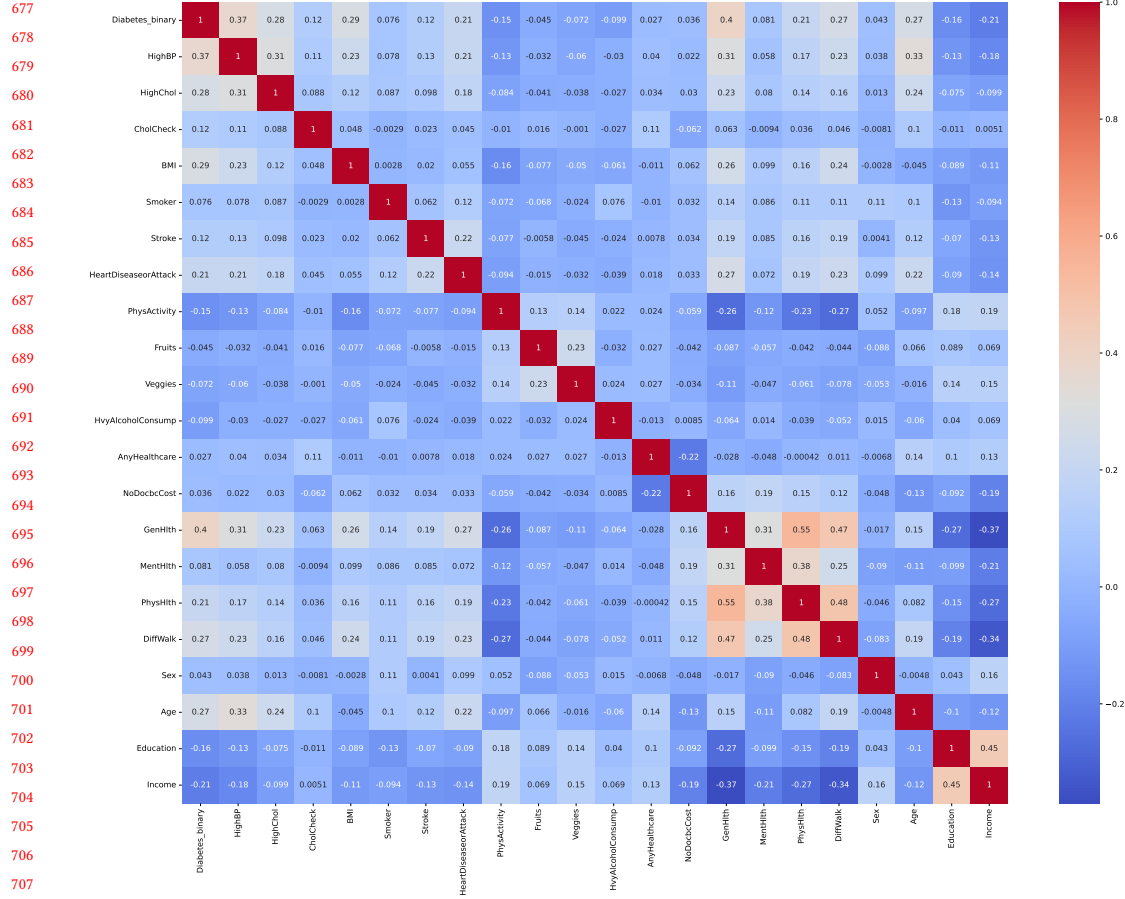


Fig. 1. Correlation Matrix of Dataset 3

Walking, and Heart Disease or Attack—emerging as powerful predictors. This suggests that clustered FL based on meaningful health indicators can both preserve data locality and maintain predictive performance.

4.6 Local Model Performances of Clusters on Dataset 3

Table 9 and Table ?? present the performance of local models trained on clusters generated from Dataset 3 based on selected features. LR and RFC were evaluated on each cluster, with accuracy, precision, recall, and f1-score reported.

For Age, two clusters were formed: cluster 0 with 47,536 records (mean age 65–69) and cluster 1 with 21,521 records (mean age 40–44). In the younger cluster (0), LR achieved a high recall of 0.8506, demonstrating strong ability to identify positive cases, while RFC provided a more balanced performance, with an f1-score of 0.7678. Cluster 1 (older group) showed higher overall accuracy (0.8038 for LR, 0.8007 for RFC) but lower recall, indicating that older participants were harder to classify correctly as diabetic.

BMI-based clusters displayed clear predictive patterns. Cluster sizes were: Underweight (<18.5) 649, Healthy Weight (18.5–24.9) 13,783, Overweight (25–29.9) 23,400, Class 1 Obesity (30–34.9) 17,111, Class 2 Obesity (35–39.9) 8,083, and

Table 8. Performance of Global Models and Aggregated FL Models for Diabetes Classification (Dataset 3)

Feature	Model	Metrics			
		Accuracy	Precision	Recall	f1-score
Global	LR	0.7405	0.7313	0.7816	0.7556
	RFC Classifier	0.7482	0.7407	0.7837	0.7616
Age	LR	0.7534	0.7357	0.8106	0.7714
	RFC Classifier	0.7442	0.7448	0.7632	0.7539
BMI	LR	0.7547	0.7330	0.8213	0.8258
	RFC Classifier	0.7397	0.7313	0.7791	0.7545
Difficulty Walking	LR	0.7525	0.7429	0.7916	0.7665
	RFC Classifier	0.7477	0.7286	0.8101	0.7672
PhysHlth	LR	0.7507	0.7476	0.7766	0.7618
	RFC Classifier	0.7492	0.7400	0.7884	0.7634
GenHlth	LR	0.4992	0.5129	0.4809	0.4964
	RFC Classifier	0.7344	0.7116	0.8115	0.7583
Heart Disease or Attack	LR	0.7504	0.7428	0.7859	0.7637
	RFC Classifier	0.7339	0.6839	0.8950	0.7754
High Cholesterol	LR	0.6756	0.6368	0.8564	0.7305
	RFC Classifier	0.7475	0.7416	0.7799	0.7603
High Blood Pressure	LR	0.6480	0.6127	0.8540	0.7135
	RFC Classifier	0.7373	0.7411	0.7505	0.7457

Class 3 Obesity (≥ 40) 6,031. LR performed best in higher BMI categories, reaching a recall of 0.9885 and f1-score of 0.8843 in Class 3 Obesity. RFC also achieved strong f1-scores above 0.80 in obese clusters. Lower BMI clusters produced weaker results, reflecting smaller sample sizes and weaker association with diabetes.

For Difficulty Walking, cluster 0 included 51,198 records and cluster 1 had 17,859 records. Both classifiers performed substantially better in the “Yes” cluster (1), with recall above 0.92, highlighting the strong link between mobility limitations and diabetes.

PhysHlth clusters contained 55,723 records in cluster 0 (mean 1.30) and 12,334 records in cluster 1 (mean 25.31). Individuals reporting poor physical health (cluster 1) showed higher recall and f1-scores (above 0.84) than those with no limitations.

GenHlth clusters had 7,607 records in “Poor,” 19,105 in “Fair,” 23,246 in “Good,” 13,292 in “Very Good,” and 5,807 in “Excellent.” LR gave unstable results with low accuracy in “Poor” and “Excellent” clusters, whereas RFC maintained strong f1-scores, particularly in the “Excellent” category (0.8623), demonstrating robustness in handling self-reported health variables.

For Heart Disease or Attack, clusters contained 58,638 records in the “No” group and 10,419 in the “Yes” group. RFC achieved a recall of 0.9604 and f1-score of 0.8700 in the “Yes” cluster, confirming the close link between cardiovascular conditions and diabetes.

High Cholesterol clusters included 32,365 records for “No” and 36,692 for “Yes,” while High Blood Pressure clusters had 29,610 records for “No” and 39,447 for “Yes.” Both features were more predictive in the “Yes” clusters, with f1-scores above 0.79. LR often reached recall above 0.94 in “No” clusters, but this came with lower precision, indicating a trade-off.

Table 9. Performance of Local Models on Clusters for Diabetes Classification (Dataset 3)

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
Age	0	LR	0.7309	0.7367	0.8506	0.7896
		RFC Classifier	0.7190	0.7536	0.7825	0.7678
	1	LR	0.8038	0.7304	0.6511	0.6885
		RFC Classifier	0.8007	0.7068	0.6863	0.6964
BMI	Underweight (<18.5)	LR	0.7805	0.6250	0.4545	0.5263
		RFC Classifier	0.7642	0.5417	0.7879	0.6420
	Healthy Weight (18.5-24.9)	LR	0.7840	0.6165	0.5613	0.5876
		RFC Classifier	0.7537	0.5390	0.7009	0.6094
	Overweight (25-29.9)	LR	0.7249	0.6870	0.7315	0.7085
		RFC Classifier	0.7261	0.6778	0.7641	0.7184
	Class 1 Obesity (30-34.9)	LR	0.7441	0.7518	0.8663	0.8050
		RFC Classifier	0.7316	0.7783	0.7828	0.7805
	Class 2 Obesity (35-39.9)	LR	0.7769	0.7790	0.9521	0.8569
		RFC Classifier	0.7341	0.8209	0.7944	0.8074
	Class 3 Obesity (>=40)	LR	0.7994	0.8000	0.9885	0.8843
		RFC Classifier	0.7872	0.8738	0.8482	0.8608
Difficulty Walking	No	LR	0.7450	0.7017	0.7126	0.7071
		RFC Classifier	0.7381	0.6825	0.7362	0.7083
	Yes	LR	0.7743	0.8041	0.9244	0.8601
		RFC Classifier	0.7757	0.8003	0.9342	0.8621
PhysHlth	0	LR	0.7483	0.7320	0.7290	0.7305
		RFC Classifier	0.7463	0.7225	0.7433	0.7327
	1	LR	0.7609	0.7853	0.9117	0.8438
		RFC Classifier	0.7617	0.7837	0.9166	0.8449
GenHlth	Poor	LR	0.2249	0.1557	1.0000	0.2695
		RFC Classifier	0.8197	0.4159	0.6468	0.5063
	Fair	LR	0.5851	0.4366	0.9245	0.5931
		RFC Classifier	0.7112	0.5446	0.7130	0.6176
	Good	LR	0.6725	0.7687	0.6272	0.6908
		RFC Classifier	0.7034	0.7248	0.7921	0.7570
	Very Good	LR	0.3474	0.8924	0.1560	0.2656
		RFC Classifier	0.7579	0.7248	0.8750	0.8454
	Excellent	LR	0.2198	0.8125	0.0141	0.0277
		RFC Classifier	0.7716	0.8220	0.9067	0.8623

Overall, these results show that feature-based clustering allows local models to capture subgroup-specific patterns more effectively than global models. Features such as BMI, Difficulty Walking, Heart Disease, and High Blood Pressure were particularly discriminative, underscoring their value for stratified analysis in federated learning scenarios.

5 Conclusion and Discussion

In conclusion, the experiments demonstrate that federated learning with feature-based clustering can achieve predictive performance comparable to centralized global models while preserving data privacy. RFC consistently outperformed LR, particularly in heterogeneous or imbalanced clusters, capturing non-linear patterns effectively. Aggregated FL models

Table 10. Performance of Local Models on Clusters for Diabetes Classification (Dataset 3) ctd.

Feature	Cluster	Model	Metrics			
			Accuracy	Precision	Recall	f1-score
Heart Disease or Attack	No	LR	0.7482	0.7225	0.7471	0.7346
		RFC Classifier	0.7256	0.6537	0.8754	0.7485
	Yes	LR	0.7626	0.8040	0.9147	0.8558
		RFC Classifier	0.7790	0.7952	0.9604	0.8700
High Cholesterol	No	LR	0.6128	0.4862	0.9575	0.6450
		RFC Classifier	0.7634	0.6689	0.7044	0.6862
	Yes	LR	0.7311	0.7823	0.8053	0.7936
		RFC Classifier	0.7335	0.7783	0.8181	0.7977
High Blood Pressure	No	LR	0.5403	0.3834	0.9496	0.5462
		RFC Classifier	0.7657	0.5981	0.5964	0.5973
	Yes	LR	0.7289	0.7878	0.8232	0.8051
		RFC Classifier	0.7160	0.7861	0.8001	0.7930

provided balanced performance across clusters, mitigating issues arising from small or single-class subgroups. The results also highlight the importance of cluster size, feature relevance, and class distribution in federated settings.

Overall, these results demonstrate that feature-based federated learning can preserve local specificity while achieving reliable global classification performance, highlighting the potential of cluster-informed FL strategies in clinical prediction tasks.

Acknowledgments

References

- [1] Rodolfo Stoffel Antunes, Cristiano André da Costa, Arne Küderle, Imrana Abdullahi Yari, and Björn Eskofier. 2022. Federated learning for healthcare: Systematic review and architecture proposal. *ACM Transactions on Intelligent Systems and Technology (TIST)* 13, 4 (2022), 1–23.
- [2] Eda Bahar and Ozgun Pinarer. 2024. Federated Learning and Resource-Constrained Embedded Systems: A Comprehensive Survey. *Transactions on Computer Science and Applications* 1, 2 (2024), 40–55.
- [3] Ali Kashif Bashir, Nancy Victor, Sweta Bhattacharya, Thien Huynh-The, Rajeswari Chengoden, Gokul Yenduri, Praveen Kumar Reddy Maddikunta, Quoc-Viet Pham, Thippa Reddy Gadekallu, and Madhusanka Liyanage. 2023. Federated learning for the healthcare metaverse: Concepts, applications, challenges, and future directions. *IEEE Internet of Things Journal* 10, 24 (2023), 21873–21891.
- [4] Theodora S Brisimi, Ruidi Chen, Theofanie Mela, Alex Olshevsky, Ioannis Ch Paschalidis, and Wei Shi. 2018. Federated learning of predictive models from federated electronic health records. *International journal of medical informatics* 112 (2018), 59–67.
- [5] Ahmad Chaddad, Yihang Wu, and Christian Desrosiers. 2023. Federated learning for healthcare applications. *IEEE internet of things journal* 11, 5 (2023), 7339–7358.
- [6] Muhammad Asaad Cheema, Muhammad Zohaib Sarwar, Daniel Cantero, and Pierluigi Salvo Rossi. 2025. Clustered federated learning for population-based structural health monitoring. *IEEE Internet of Things Journal* (2025).
- [7] Francesco Cremonesi, Vincent Planat, Varvara Kalokyri, Haridimos Kondylakis, Tiziana Sanavia, Victor Miguel Mateos Resinas, Babita Singh, and Silvia Uribe. 2023. The need for multimodal health data modeling: A practical approach for a federated-learning healthcare platform. *Journal of biomedical informatics* 141 (2023), 104338.
- [8] Trung Kien Dang, Xiang Lan, Jianshu Weng, and Mengling Feng. 2022. Federated learning for electronic health records. *ACM Transactions on Intelligent Systems and Technology (TIST)* 13, 5 (2022), 1–17.
- [9] Ittai Dayan, Holger R Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z Abidin, Andrew Liu, Anthony Beardsworth Costa, Bradford J Wood, Chien-Sung Tsai, et al. 2021. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature medicine* 27, 10 (2021), 1735–1743.
- [10] Gaurav Dhiman, Sapna Juneja, Hamidreza Mohafez, Ibrahim El-Bayoumy, Lokesh Kumar Sharma, Maryam Hadizadeh, Mohammad Aminul Islam, Wattana Viriyasitavat, and Mayeen Uddin Khandaker. 2022. Federated learning approach to protect healthcare data over big data scenario. *Sustainability* 14, 5 (2022), 2500.

- [11] Sunil R Dommaraju, Stephanie Gordon Rivera, Ethan G Rocha, Scott Bicknell, Daniel Loizzo, Ayesha Mohammad, Priya Rajan, Alexandria Seballos, Avisek Datta, Rashid Ahmed, et al. 2022. Health professional students at the University of Illinois Chicago (HOLISTIC) Cohort Study: A protocol. *Plos one* 17, 8 (2022), e0269964.
- [12] Haya Elayan, Moayad Aloqaily, and Mohsen Guizani. 2021. Sustainability of healthcare data analysis IoT-based systems using deep federated learning. *IEEE Internet of Things Journal* 9, 10 (2021), 7338–7346.
- [13] Kehua Guo, Tianyu Chen, Sheng Ren, Nan Li, Min Hu, and Jian Kang. 2022. Federated learning empowered real-time medical data processing method for smart healthcare. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 21, 4 (2022), 869–879.
- [14] Mohammad Hossin and Md Nasir Sulaiman. 2015. A review on evaluation metrics for data classification evaluations. *International journal of data mining & knowledge management process* 5, 2 (2015), 1.
- [15] Shiyi Jiang, Farshad Firouzi, and Krishnendu Chakraborty. 2023. Low-overhead clustered federated learning for personalized stress monitoring. *IEEE Internet of Things Journal* 11, 3 (2023), 4335–4347.
- [16] Madhura Joshi, Ankit Pal, and Malaikannan Sankarasubbu. 2022. Federated learning for healthcare domain-pipeline, applications and challenges. *ACM Transactions on Computing for Healthcare* 3, 4 (2022), 1–36.
- [17] Dasaradham Reddy Kandati and Thippa Reddy Gadekallu. 2022. Genetic clustered federated learning for COVID-19 detection. *Electronics* 11, 17 (2022), 2714.
- [18] Dinh C Nguyen, Quoc-Viet Pham, Pubudu N Pathirana, Ming Ding, Aruna Seneviratne, Zihuai Lin, Octavia Dobre, and Won-Joo Hwang. 2022. Federated learning for smart healthcare: A survey. *ACM Computing Surveys (Csur)* 55, 3 (2022), 1–37.
- [19] Godwin Ogbuabor and FN Ugwoke. 2018. Clustering algorithm for a healthcare dataset using silhouette score value. *Int. J. Comput. Sci. Inf. Technol* 10, 2 (2018), 27–37.
- [20] Bjarne Pfitzner, Nico Steckhan, and Bert Arnrich. 2021. Federated learning in a medical context: a systematic literature review. *ACM Transactions on Internet Technology (TOIT)* 21, 2 (2021), 1–31.
- [21] Prayitno, Chi-Ren Shyu, Karisma Trinanda Putra, Hsing-Chung Chen, Yuan-Yu Tsai, KSM Tozammel Hossain, Wei Jiang, and Zon-Yin Shae. 2021. A systematic review of federated learning in the healthcare area: From the perspective of data properties and applications. *Applied Sciences* 11, 23 (2021), 11191.
- [22] Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletari, Holger R Roth, Shadi Albarqouni, Spyridon Bakas, Mathieu N Galtier, Bennett A Landman, Klaus Maier-Hein, et al. 2020. The future of digital health with federated learning. *NPJ digital medicine* 3, 1 (2020), 119.
- [23] Shariq I Sherwani, Haseeb A Khan, Aishah Ekhzaimy, Afshan Masood, and Meena K Sakharkar. 2016. Significance of HbA1c test in diagnosis and prognosis of diabetic patients. *Biomarker insights* 11 (2016), BMI-S38440.
- [24] Ischa Stranders, Michaela Diamant, Rogier E van Gelder, Hugo J Spruijt, Jos WR Twisk, Robert J Heine, and Frans C Visser. 2004. Admission blood glucose level as risk indicator of death after myocardial infarction in patients with and without diabetes mellitus. *Archives of internal medicine* 164, 9 (2004), 982–988.
- [25] Le Sun and Jin Wu. 2022. A scalable and transferable federated learning system for classifying healthcare sensor data. *IEEE journal of biomedical and health informatics* 27, 2 (2022), 866–877.
- [26] W Timothy Garvey. 2018. Clinical definition of overweight and obesity. In *Bariatric endocrinology: evaluation and management of adiposity, Adiposopathy and related diseases*. Springer, 121–143.
- [27] Jie Xu, Benjamin S Glicksberg, Chang Su, Peter Walker, Jiang Bian, and Fei Wang. 2021. Federated learning for healthcare informatics. *Journal of healthcare informatics research* 5, 1 (2021), 1–19.
- [28] Xiaohang Xu, Hao Peng, Lichao Sun, Md Zakirul Alam Bhuiyan, Lianzhong Liu, and Lifang He. 2021. Fedmood: Federated learning on mobile health data for mood detection. *arXiv preprint arXiv:2102.09342* (2021).
- [29] Wangyouchen Zhang, Zhenhua Xia, Guoqing Cai, Junhao Wang, and Xutao Dong. 2025. Enhancing diabetes risk prediction through focal active learning and machine learning models. *PLoS One* 20, 7 (2025), e0327120.
- [30] Xiaokang Zhou, Xiaozhou Ye, I Kevin, Kai Wang, Wei Liang, Nirmal Kumar C Nair, Shohei Shimizu, Zheng Yan, and Qun Jin. 2023. Hierarchical federated learning with social context clustering-based participant selection for internet of medical things applications. *IEEE Transactions on Computational Social Systems* 10, 4 (2023), 1742–1751.